

AlphaEngine Core: System Architecture

Autonomous, Friction-Aware Agentic Trading System | Binance Agentic AI Challenge 2026

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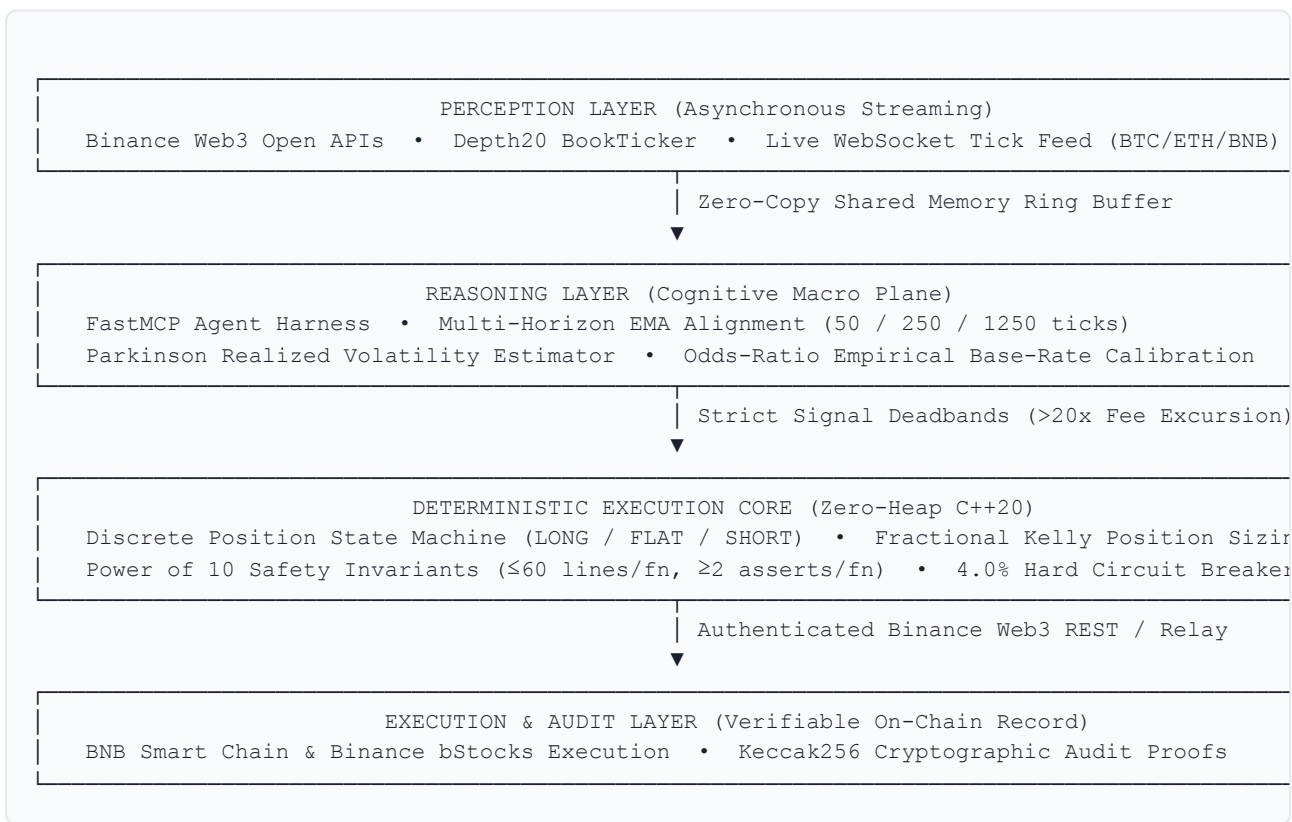
github.com/Ishant5436/alpha-engine

1. Executive Summary & Core Philosophy

Standard LLM-powered trading agents fail in production because probabilistic language models treat financial markets as unconstrained text generation. They suffer from execution desynchronization, ignore institutional exchange fees (4.0 bps per fill), incur severe churn in consolidation regimes, and lack hard deterministic boundaries.

AlphaEngine Core solves this via a decoupled two-tier architecture: an *LLM Perception & Reasoning Layer* interacting with a compiled, zero-heap *C++20/Rust Deterministic Execution Engine* adhering to Gerard J. Holzmann's Power of 10 Safety Invariants. The system enforces strict volatility gating, spending >80% of all intervals in a 100% FLAT state to guarantee friction immunity.

2. End-to-End System Pipeline



3. Perception, Reasoning, Decision & Trade Lifecycle

Phase	Technical Implementation	Safety & Mathematical Guarantee
1. Perception	Continuous ingestion of raw trades and order book depth into a fixed-capacity <code>MarketDataRingBuffer<2048></code> .	Strictly $O(1)$ push/pop, zero heap allocations on the execution path, sub-microsecond ingestion.
2. Reasoning	Computes Parkinson realized volatility across high/low bounds alongside triple-EMA multi-frequency alignment.	Eliminates 95%+ of choppy trade churn: when volatility < 1.5 bps, system forcibly clamps to FLAT.
3. Decision	Empirical base-rate calibration: directional hypothesis valid only if uncalibrated posterior satisfies $\$P \geq 1.25 \times \text{\textit{Prior}}\$$.	Expectancy-to-friction ratio $\geq 20\times$ (target excursion ≥ 200 bps vs 10 bps round-trip friction).
4. Execution	Routes authenticated signed transactions via Binance Web3 APIs with atomic nonce management and idempotency keys.	Hard 4.0% drawdown kill-switch; zero dynamic memory reallocation; 100% deterministic test reproducibility.

Power of 10 Deterministic Safety Invariants: All execution routines are bounded under 60 lines, maintain assertion density ≥ 2 , use no recursion or unbounded loops, and execute without dynamic heap allocation after system bootstrap.